

25/04/2025



# Aakash

Medical | IIT-JEE | Foundations

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CODE-A



## AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

### CST-12

Time : 180 Mins.

### Answers

|         |         |          |          |          |
|---------|---------|----------|----------|----------|
| 1. (2)  | 37. (3) | 73. (4)  | 109. (2) | 145. (2) |
| 2. (2)  | 38. (4) | 74. (3)  | 110. (2) | 146. (4) |
| 3. (3)  | 39. (3) | 75. (4)  | 111. (1) | 147. (2) |
| 4. (1)  | 40. (2) | 76. (1)  | 112. (3) | 148. (1) |
| 5. (2)  | 41. (4) | 77. (2)  | 113. (3) | 149. (2) |
| 6. (3)  | 42. (2) | 78. (4)  | 114. (3) | 150. (3) |
| 7. (1)  | 43. (2) | 79. (3)  | 115. (3) | 151. (2) |
| 8. (2)  | 44. (2) | 80. (4)  | 116. (2) | 152. (2) |
| 9. (4)  | 45. (4) | 81. (3)  | 117. (3) | 153. (3) |
| 10. (4) | 46. (3) | 82. (3)  | 118. (4) | 154. (3) |
| 11. (2) | 47. (2) | 83. (1)  | 119. (2) | 155. (2) |
| 12. (2) | 48. (4) | 84. (1)  | 120. (1) | 156. (3) |
| 13. (4) | 49. (2) | 85. (1)  | 121. (3) | 157. (2) |
| 14. (3) | 50. (2) | 86. (2)  | 122. (4) | 158. (2) |
| 15. (1) | 51. (4) | 87. (3)  | 123. (3) | 159. (1) |
| 16. (2) | 52. (3) | 88. (1)  | 124. (3) | 160. (2) |
| 17. (1) | 53. (2) | 89. (3)  | 125. (2) | 161. (3) |
| 18. (2) | 54. (2) | 90. (2)  | 126. (4) | 162. (1) |
| 19. (1) | 55. (3) | 91. (2)  | 127. (1) | 163. (2) |
| 20. (2) | 56. (3) | 92. (1)  | 128. (2) | 164. (2) |
| 21. (3) | 57. (3) | 93. (3)  | 129. (4) | 165. (2) |
| 22. (1) | 58. (3) | 94. (3)  | 130. (1) | 166. (2) |
| 23. (4) | 59. (2) | 95. (1)  | 131. (1) | 167. (3) |
| 24. (1) | 60. (2) | 96. (4)  | 132. (1) | 168. (3) |
| 25. (3) | 61. (3) | 97. (4)  | 133. (2) | 169. (3) |
| 26. (2) | 62. (1) | 98. (2)  | 134. (1) | 170. (4) |
| 27. (1) | 63. (2) | 99. (2)  | 135. (4) | 171. (1) |
| 28. (2) | 64. (3) | 100. (3) | 136. (1) | 172. (1) |
| 29. (2) | 65. (4) | 101. (3) | 137. (1) | 173. (2) |
| 30. (2) | 66. (1) | 102. (3) | 138. (4) | 174. (3) |
| 31. (2) | 67. (4) | 103. (1) | 139. (3) | 175. (2) |
| 32. (4) | 68. (2) | 104. (1) | 140. (1) | 176. (2) |
| 33. (4) | 69. (2) | 105. (2) | 141. (1) | 177. (1) |
| 34. (4) | 70. (3) | 106. (2) | 142. (4) | 178. (1) |
| 35. (2) | 71. (4) | 107. (4) | 143. (1) | 179. (4) |
| 36. (4) | 72. (4) | 108. (1) | 144. (3) | 180. (3) |

## Hints and Solutions

## PHYSICS

(1) Answer : (2)

Solution:

Absolute error = |Measurement – Mean value|

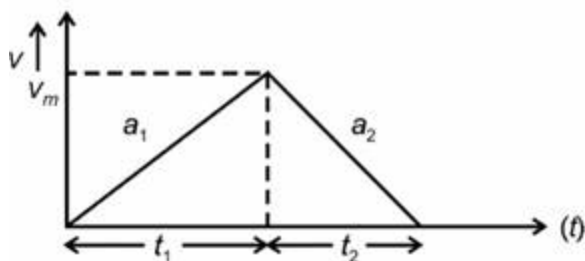
Hence, absolute error has the same unit as the quantity itself.

$$\text{Fractional error} = \frac{\text{Mean absolute error}}{\text{Mean value of measurement}}$$

So, fractional error has no unit

(2) Answer : (2)

Solution:



$$a_1 = 2 \text{ m s}^{-2}, a_2 = 3 \text{ m s}^{-2}$$

 $T = 40 \text{ s}$  (total time)

$$a_1 = \frac{v_m}{t_1}, a_2 = \frac{v_m}{t_2}$$

$$T = t_1 + t_2 = \frac{v_m}{a_1} + \frac{v_m}{a_2}$$

$$v_m = \frac{a_1 \times a_2}{(a_1 + a_2)} T$$

$$\therefore v_m = \frac{2 \times 3}{2 + 3} \times 40 = \frac{6}{5} \times 40 = 48 \text{ m/s}$$

(3) Answer : (3)

Solution:

For a hydraulic lift:

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow \frac{F_1}{\pi(10)^2} = \frac{1500 \times 10}{\pi(20)^2}$$

$$F_1 = \frac{1500 \times 10 \times 100}{400} = 3.75 \times 10^3 \text{ N}$$

(4) Answer : (1)

Solution:

 Ideal gas equation:  $PV = nRT = \frac{N}{N_0}RT$ 

$$\frac{N_B}{N_A} = \frac{P_B V_B}{T_B} \times \frac{T_A}{P_A V_A}$$

$$\frac{N_B}{N_A} = \frac{2P \times \frac{V}{2}}{2T} \times \frac{T}{PV} = \frac{1}{2}$$

(5) Answer : (2)

Solution:

Applying equation of continuity:

$$A_1 v_1 = A_2 v_2$$

 If  $A_2 < A_1$  then  $v_2 > v_1$ 

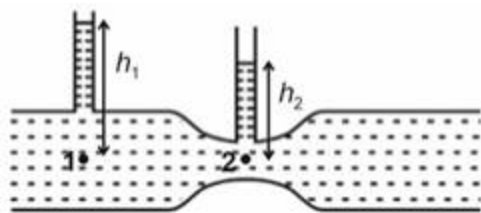
Applying Bernoulli's equation:

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) > 0$$

$$\therefore P_1 > P_2 \Rightarrow \rho g h_1 > \rho g h_2$$

$$\therefore h_1 > h_2$$



(6) Answer : (3)

**Solution:**

In logic gates, we get output of inputs based on some relation or logic.

NOR and NAND gates are considered as universal gates.

(7) Answer : (1)

**Solution:**

$$\Delta M = [Zm_p + (A - Z)m_n] - M$$

$$\Delta M = [(2 \times 1.0073) + (4 - 2) \times 1.0087] - 4.0015$$

$$\Delta M = 0.0305 \text{ u}$$

Energy equivalent of 1 u = 931.5 MeV

$$\therefore E_b = 0.0305 \times 931.5 \text{ MeV}$$

$$E_b = 28.4 \text{ MeV}$$

(8) Answer : (2)

**Solution:**

$$P_{av} = v_{rms} \cdot I_{rms} \cdot \cos \phi$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$$

$$\text{Power factor, } \cos \phi = \frac{R}{Z}$$

(9) Answer : (4)

**Solution:**

Intensity of magnetisation

$$I = \frac{M}{1A} = \frac{3}{3 \times 10^{-2} \times 2 \times 10^{-4}}$$

$$= 5 \times 10^5 \text{ A/m}$$

(10) Answer : (4)

**Solution:**

In thermodynamics, heat and work can be converted into one another.

Heat and work are path variables.

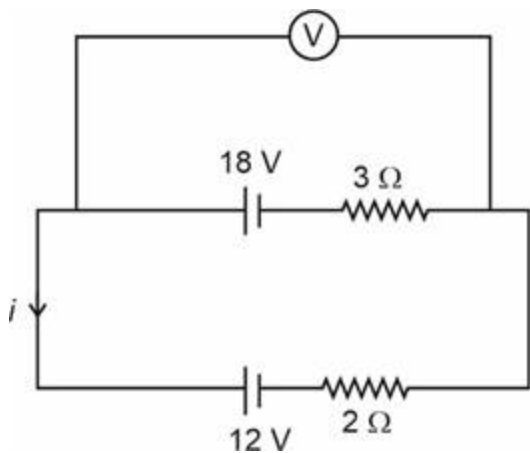
(11) Answer : (2)

**Solution:**

|           |   |                                                                                |
|-----------|---|--------------------------------------------------------------------------------|
| Infra-red | – | Vibration of atoms and molecules                                               |
| Radio     | – | Rapid vibration of electrons in aerials                                        |
| Light     | – | Electrons in atoms emit light when they move from higher to lower energy level |
| Microwave | – | Klystron valve                                                                 |

(12) Answer : (2)

**Solution:**



$$i = \frac{V}{R} = \frac{18-12}{3+2} = \frac{6}{5} \text{ A}$$

$$V = \varepsilon - ir \Rightarrow V = 18 - \frac{6}{5} \times 3 = 14.4 \text{ V}$$

(13) Answer : (4)

**Solution:**

Work done by gas in adiabatic process:

$$W = \frac{P_i V_i - P_f V_f}{\gamma - 1}$$

using  $PV = nRT$

$$P_i V_i = 2R \times 300 = 600R$$

$$P_f V_f = 2R \times 600 = 1200R$$

$$\gamma = \frac{5}{3} \text{ for monoatomic}$$

$$W = \frac{600R - 1200R}{\frac{5}{3} - 1} = \frac{-600R \times 3}{2} = -900R$$

(14) Answer : (3)

**Solution:**

In a stress vs strain graph, slope gives Young's modulus.

$$Y_A = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$Y_B = \tan 60^\circ = \sqrt{3}$$

$$\frac{Y_A}{Y_B} = \frac{\frac{1}{\sqrt{3}}}{\sqrt{3}} \Rightarrow \frac{Y}{Y_B} = \frac{1}{3} \Rightarrow Y_B = 3Y$$

(15) Answer : (1)

**Solution:**

$$I = \frac{MR^2}{2} = \frac{2 \times 1^2}{2} = 1 \text{ kg m}^2$$

$$\tau = I\alpha$$

$$4 = 1 \times \alpha \Rightarrow \alpha = 4 \text{ rad/s}^2$$

$$\omega = \omega_0 + \alpha t \Rightarrow \omega = 0 + 4 \times 4 = 16 \text{ rad/s}$$

(16) Answer : (2)

**Solution:**

For an orbiting satellite:

$$TE = -KE = \frac{PE}{2}$$

(17) Answer : (1)

**Solution:**

$$\vec{V}_{S/R} = \vec{v}_S - \vec{v}_R$$

$$\vec{V}_S = \vec{v}_{S/R} + \vec{v}_R$$

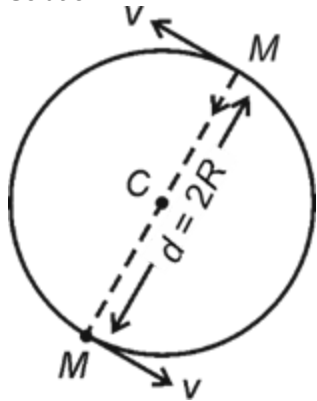
$$V_s = \sqrt{8^2 + 4^2}$$

$$= \sqrt{80}$$

$$\simeq 9 \text{ m/s}$$

(18) Answer : (2)

Solution:



$$F = \frac{G \cdot M \cdot M}{d^2} = \frac{GM^2}{(2R)^2} = \frac{GM^2}{4R^2}$$

(19) Answer : (1)

Solution:

$$y = 10\sin(2x)\cos(2t)$$

For node at time  $t$

$$10\sin 2x = 0$$

$$2x = 0, \pi, 2\pi$$

$$2x = \pi$$

$$x = \frac{\pi}{2}$$

(20) Answer : (2)

Solution:

$$v = \sqrt{\frac{T}{\mu}}, \quad \mu = \frac{m}{l}$$

$$v = \sqrt{\frac{Tl}{m}} = \sqrt{\frac{80 \times 0.5}{4 \times 10^{-3}}} = \sqrt{\frac{40 \times 10^3}{4}} = 100 \text{ m/s}$$

(21) Answer : (3)

Solution:

$$\text{Width of central bright fringe} = 2(0.8) = 1.6 \text{ mm}$$

(22) Answer : (1)

Solution:

$$P = I \times A = 1400 \times 8 \times 20 = 224 \times 10^3 \text{ W}$$

$$= 224 \text{ kW}$$

(23) Answer : (4)

Solution:

From Gauss's law

$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

Here  $E$  is net electric field due to all charges

(24) Answer : (1)

Solution:

$$\text{Charge stored, } q_A = CV$$

$$q_B = KCV$$

$$q_C = 2KCV$$

therefore,

$$q_A : q_B : q_C = 1 : K : 2K$$

(25) Answer : (3)

Solution:

$$Q = \text{Area under } I - t \text{ graph}$$

$$= 30 + 30 = 60 \text{ C}$$

(26) Answer : (2)

Solution:

$$V_O - V_Q = \frac{1}{2} B \omega (3L)^2 \dots (i)$$

$$V_O - V_P = \frac{1}{2} B \omega L^2 \dots (ii)$$

Subtracting equation (i) – (ii)

$$V_P - V_Q = 4B\omega L^2$$

(27) Answer : (1)

Solution:

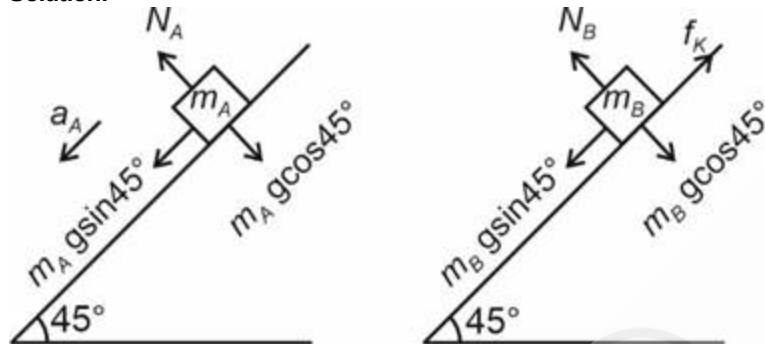
$$E_{\min} = \frac{hc}{\lambda_{\max}} \text{ so } m = 2$$

$$E_{\min} = \frac{12400}{1800 \times \frac{4}{3}} \text{ eV}$$

$$\simeq 5.2 \text{ eV}$$

(28) Answer : (2)

Solution:



$$m_A g \sin 45^\circ = m_A a_A$$

$$a_A = g \sin 45^\circ = \frac{g}{\sqrt{2}}$$

$$m_B g \sin 45^\circ - f_K = m_B a_B$$

$$f_K = \mu \times m_B g \cos 45^\circ$$

$$m_B g \sin 45^\circ - \frac{1}{2} \times m_B g \cos 45^\circ = m_B a_B$$

$$a_B = \frac{g}{\sqrt{2}} - \frac{g}{2\sqrt{2}} = \frac{g}{2\sqrt{2}}$$

$$\therefore \frac{a_A}{a_B} = \frac{\frac{g}{\sqrt{2}}}{\frac{g}{2\sqrt{2}}} = 2$$

(29) Answer : (2)

Solution:

From conservation of energy, we get

$$v_B = \sqrt{3gl} \text{ and } v_C = \sqrt{gl}$$

$$\therefore \frac{KE_B}{KE_C} = \frac{v_B^2}{v_C^2} = \frac{3}{1}$$

(30) Answer : (2)

Solution:

$$a_{\text{cm}} = \frac{m_1 a_1 + m_2 a_2}{m_1 + m_2} = \frac{m \times 0 + m \times a}{2m}$$

$$a_{\text{cm}} = \frac{a}{2}$$

(31) Answer : (2)

Solution:

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 3 & 5 \\ 1 & 2 & 3 \end{vmatrix}$$

$$\vec{\tau} = \hat{i}(9 - 10) - \hat{j}(21 - 5) + \hat{k}(14 - 3)$$

$$\vec{\tau} = (-\hat{i} - 16\hat{j} + 11\hat{k}) \text{ N m}$$

(32) Answer : (4)

**Solution:**

Rotation of earth about its own axis is a periodic motion but not SHM.

(33) Answer : (4)

**Solution:**

ice  $\xrightarrow{Q_1}$  water  
0°C      0°C

$$Q_1 = mL_f = \frac{150}{1000} \times 3.34 \times 10^5 = 50.1 \text{ kJ}$$

water  $\xrightarrow{Q_2}$  water  
50°C      0°C

$$Q_2 = ms\Delta T = \frac{60}{1000} \times 4186 \times 50 = 12.558 \text{ kJ}$$

Since  $Q_1 > Q_2$ , ice will not melt completely, hence temperature will be 0°C.

(34) Answer : (4)

**Solution:**

$\beta^+$  decay:  ${}_1^1p \rightarrow {}_0^1n + {}_1^0e + \nu$

(35) Answer : (2)

**Solution:**

Dynamic resistance,  $R_D = \frac{\Delta V}{\Delta I}$

$$R_D = \frac{1.2-1}{(20-10) \times 10^{-3}} = \frac{0.2 \times 10^3}{10} = 20 \Omega$$

(36) Answer : (4)

**Solution:**

$$E = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$$

$$\text{For hydrogen atom} \Rightarrow E = \frac{-13.6 \times 1}{n^2} \text{ eV} \dots (i)$$

$$\text{For He}^+ \text{ ion} \Rightarrow E' = \frac{-13.6 \times 2^2}{n^2} \text{ eV} \dots (ii)$$

$$\frac{E}{E'} = \frac{1}{4} \Rightarrow E' = 4E$$

(37) Answer : (3)

**Solution:**

$f = 12 \text{ cm}$ ,  $R_1 = 15 \text{ cm}$ ,  $R_2 = 10 \text{ cm}$

$$\frac{1}{12} = (\mu - 1) \left[ \frac{1}{15} + \frac{1}{10} \right] = (\mu - 1) \left[ \frac{2+3}{30} \right] = \frac{\mu - 1}{6}$$

$$(\mu - 1) = 0.5 \therefore \mu = 1.5$$

(38) Answer : (4)

**Solution:**

Fringe width in YDSE is given by

$$\beta = \frac{\lambda D}{d}$$

(39) Answer : (3)

**Solution:**

Kirchhoff's junction rule is based on conservation of charge.

(40) Answer : (2)

**Solution:**

$$\vec{v} = (3\hat{i} + 4\hat{j}) \text{ m/s and } \vec{a} = (4\hat{i} - 3\hat{j}) \text{ m/s}^2$$

$$\vec{v} \cdot \vec{a} = (3\hat{i} + 4\hat{j}) \cdot (4\hat{i} - 3\hat{j})$$

$$= 12 - 12$$

$$= 0$$

$\therefore \vec{v}$  and  $\vec{a}$  are perpendicular and  $\vec{a}$  is constant.

Hence path will be parabolic.

(41) Answer : (4)

**Solution:**

In projectile motion, maximum height attained would be less than that in absence of air.

(42) Answer : (2)

**Solution:**

From, volume conservation,

$$\frac{4}{3}\pi R^3 = n \times \frac{4}{3}\pi r^3 \Rightarrow R = (n)^{\frac{1}{3}} r$$

Now, excess pressure in big drop

$$= \frac{2T}{R} = \frac{2T}{(n)^{\frac{1}{3}} r}$$

(43) Answer : (2)

**Solution:**

Maximum kinetic energy =  $eV_S = 2.5 \text{ eV}$

(44) Answer : (2)

**Solution:**

$$F = 0$$

$$15 - 3x = 0$$

$$x = 5 \text{ m}$$

(45) Answer : (4)

**Solution:**

If  $\vec{v}$ ,  $\vec{E}$  and  $\vec{B}$  are mutually perpendicular, then for no deflection of the charged particle

$$v = \frac{E}{B} = \frac{6 \times 10^4}{1.2} = 5 \times 10^4 \text{ m/s}$$

CHEMISTRY

(46) Answer : (3)

**Solution:**

|                  | C                   | H                 |
|------------------|---------------------|-------------------|
| Mass%            | 80                  | 20                |
|                  | $\frac{80}{12}$     | $\frac{20}{1}$    |
| Mole             | 6.66                | 20                |
|                  | $\frac{6.66}{6.66}$ | $\frac{20}{6.66}$ |
| Simplest ratio : | 1                   | 3                 |

Empirical formula  $\Rightarrow \text{CH}_3$

(47) Answer : (2)

**Solution:**

According to Heisenberg uncertainty principle

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \times m \Delta v \geq \frac{h}{4\pi}$$

$$\Delta x \times \Delta v \geq \frac{h}{4\pi m}$$

(48) Answer : (4)

**Solution:**

Value of  $m$  varies from '-l' to '+l' hence for  $l = 1$ ,  $m$  cannot be -2.

(49) Answer : (2)

**Solution:**

On moving left to right in a period IP increases.

•  $2s^2$  has more IP than  $2p^1$

Hence order  $\Rightarrow \text{Li} < \text{B} < \text{Be} < \text{C}$

(50) Answer : (2)

**Solution:**



Element with most metallic nature would have less electronegativity.

(51) Answer : (4)

**Solution:**

Canonical form does not have any real existence hence there is no equilibrium.

(52) Answer : (3)

**Solution:**

$\text{BrF}_5$  is square pyramidal with  $sp^3d^2$  hybridisation.

(53) Answer : (2)

**Solution:**

For reaction

$$\Delta_r H = \sum (\Delta_f H)_{\text{Product}} - \sum (\Delta_f H)_{\text{Reactant}}$$

$$\Delta_r H = -630 - 400 + 1200$$

$$= +170 \text{ kJ mol}^{-1}$$

(54) Answer : (2)

**Hint:**

$$K_C = [\text{CO}_2]$$

**Solution:**



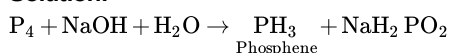
$$K_C = [\text{CO}_2]$$

$$0.5 \left( \frac{\text{mol.}}{\ell} \right) = [\text{CO}_2]$$

$$\text{moles of CO}_2 = 0.5 \times 5 = 2.5$$

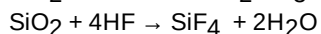
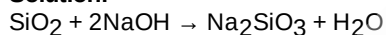
(55) Answer : (3)

**Solution:**



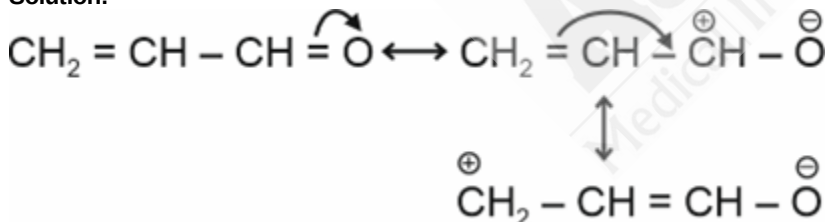
(56) Answer : (3)

**Solution:**



(57) Answer : (3)

**Solution:**



(58) Answer : (3)

**Solution:**

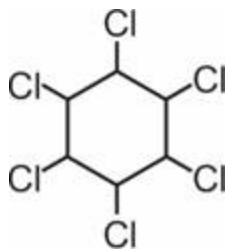
$$\% N = \frac{\text{Mass of N}}{\text{Mass of compound}} \times 100$$

$$= \frac{1 \times 2 \times 10 \times 10^{-3} \times 14}{0.4} \times 100$$

$$= 70\%$$

(59) Answer : (2)

**Solution:**



Benzene hexachloride or Gammexane.

(60) Answer : (2)

**Solution:**

$$m = \frac{X_B}{X_A} \times \frac{1000}{MWA}$$

$$= \frac{0.2}{0.8} \times \frac{1000}{18}$$

$$= 13.88 \text{ m}$$

(61) Answer : (3)

**Solution:**

In this case cell will behave like electrolytic cell and current will flow from Cu anode to Zn cathode.

(62) Answer : (1)

**Solution:**

Slowest step is rate determining step

$$r = K[X][Y_2]$$

Now

$$K_C = \frac{[X]}{[X_2]}$$

$$[X] = \sqrt{K_C(X_2)}$$

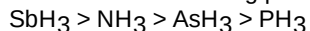
$$r = K\sqrt{K_C}(X_2)^{\frac{1}{2}}(Y_2)$$

$$r = K'(X_2)^{\frac{1}{2}}(Y_2)$$

(63) Answer : (2)

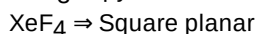
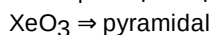
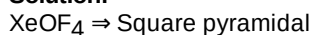
**Solution:**

Correct order of boiling point



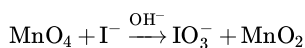
(64) Answer : (3)

**Solution:**



(65) Answer : (4)

**Solution:**



(66) Answer : (1)

**Solution:**

Zr and Hf has similar radii due to inert pair effect.

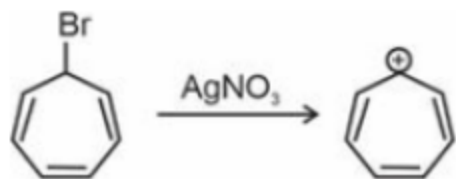
(67) Answer : (4)

**Solution:**

In an octahedral complex  $d_{x^2-y^2}$  and  $d_{z^2}$  will have higher energy than other three  $d$ -orbitals.

(68) Answer : (2)

**Solution:**

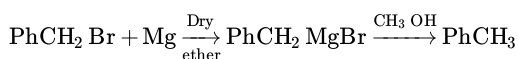


Aromatic species

Aromatic compound

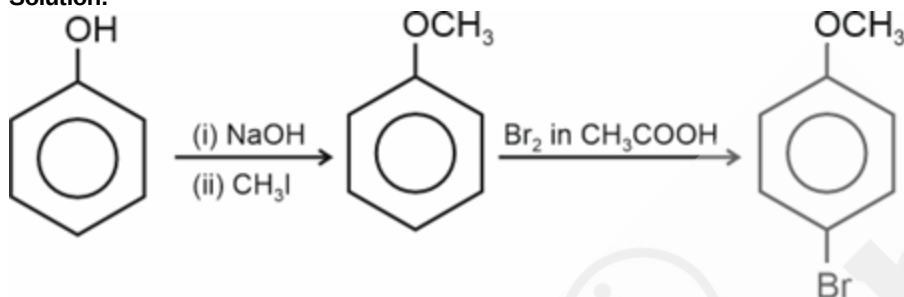
(69) Answer : (2)

Solution:



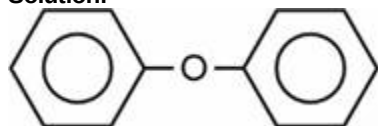
(70) Answer : (3)

Solution:



(71) Answer : (4)

Solution:



is most difficult to cleave on heating with HBr due to partial double bond character.

(72) Answer : (4)

Solution:

$2p_z$  cannot combine with  $2p_x$  or  $2p_y$  due to orientation.

(73) Answer : (4)

Solution:

For adiabatic process

$$q = 0$$

$$\text{hence } \Delta U = W$$

(74) Answer : (3)

Solution:

Mixture of  $\text{NH}_4\text{OH}$  +  $\text{NH}_4\text{Cl}$  act as basic buffer.

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]}$$

$$\text{pOH} = 14 - 10.25$$

$$= 3.75$$

$$3.75 = \text{pK}_b + \log \frac{[0.01]}{[0.1]}$$

$$3.75 = \text{pK}_b + \log [0.1]$$

$$\text{pK}_b = 3.75 - \log [0.1]$$

$$\text{pK}_b = 4.75$$

(75) Answer : (4)

Solution:

The t-butyl free radical is stabilised by greater +H and more +I.

(76) Answer : (1)

**Solution:**

Cyclic, planar species having  $(4n + 2)\pi$  electrons in conjugation will be aromatic in nature.

(77) Answer : (2)

**Solution:**

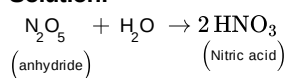
The value of the constant A depend upon the temperature and solvent.

(78) Answer : (4)

**Solution:**

For a zero order reaction  $t_{1/2} = \frac{a_0}{2k}$

(79) Answer : (3)

**Solution:**


(80) Answer : (4)

**Solution:**

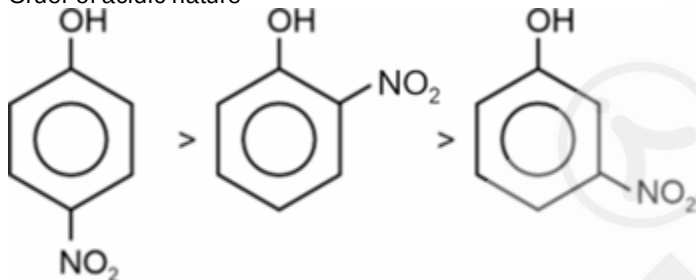
More the electron density on metal ion more will be electron donation to  $\pi^*$  orbital of CO ligand as a result C–O bond order decreases and C–O bond length increases.

(81) Answer : (3)

**Solution:**

o-nitrophenol is less acidic due to intramolecular H-bonding.

Order of acidic nature



(82) Answer : (3)

**Solution:**

Propanone has less electrophilic carbon than benzaldehyde

(83) Answer : (1)

**Solution:**

Due to +M nature of  $-\text{NH}_2$ , aniline is ortho/para directing while anilinium ion is meta directing.

(84) Answer : (1)

**Solution:**
**Vitamins Deficiency Diseases**

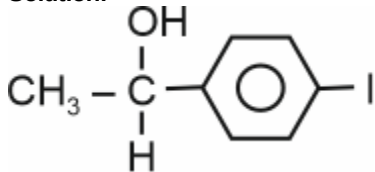
Thiamine Beri Beri

Riboflavin Cheilosis

Pyridoxine Convulsions

Ascorbic Acid Scurvy

(85) Answer : (1)

**Solution:**


can give haloform test.

(86) Answer : (2)

**Solution:**

$$p = p_A + p_B = 120 \times \frac{2}{5} + 80 \times \frac{3}{5} = 96 \text{ torr}$$

mole fraction of B in the vapour phase,

$$Y_B = \frac{p_B}{p_{total}} = \frac{80 \times \frac{3}{5}}{96} = \frac{1}{2}$$

(87) Answer : (3)

**Solution:**

Complete hydrolysis of RNA gives  $\beta$ -D-ribose, phosphoric acid and nitrogen containing heterocyclic compounds.

(88) Answer : (1)

**Solution:**

As the basic strength increases  $pK_b$  decreases order of basic strength

$(C_2H_5)_2NH > (CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$

(89) Answer : (3)

**Solution:**

| Cation    | Group | Reagent                 |
|-----------|-------|-------------------------|
| $Cu^{2+}$ | (i)   | HCl + $H_2S$            |
| $Ba^{2+}$ | (ii)  | $(NH_4)_2CO_3 + NH_4OH$ |
| $Al^{3+}$ | (iii) | $NH_4OH + NH_4Cl$       |
| $Ni^{2+}$ | (iv)  | $NH_4OH + H_2S$         |

(90) Answer : (2)

**Solution:**

Sulphide ion gives rotten egg like smell of  $H_2S(g)$  on reaction with dilute acid.

BIOLOGY

(91) Answer : (2)

**Solution:**

Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level.

(92) Answer : (1)

**Solution:**

In Deuteromycetes, sexual reproduction is absent and therefore, they are called fungi imperfecti.

(93) Answer : (3)

**Solution:**

Sporozoans lack locomotory structure. *Plasmodium* is a parasitic sporozoan, which lacks locomotory structure.

(94) Answer : (3)

**Solution:**

Bacteria as a group shows the most extensive metabolic diversity.

Bacteria may be photosynthetic autotrophic or chemosynthetic autotrophic.

(95) Answer : (1)

**Solution:**

They can be seen with naked eye e.g. bacterial colonies.

They show primitive type of sexual reproduction.

(96) Answer : (4)

**Solution:**

*Chara* is a multicellular green alga.

*Volvox* is a colonial green alga.

(97) Answer : (4)

**Solution:**

Gulmohur exhibits zygomorphic flowers with imbricate aestivation. *Datura* shows actinomorphic flower. Cotton shows twisted aestivation and *Calotropis* shows valvate aestivation.

(98) Answer : (2)

**Solution:**

*Petunia* is the ornamental plant of solanaceae family

(99) Answer : (2)

**Solution:**

In heterosporous pteridophytes, the female gametophytes are retained on parent sporophyte for variable periods.

(100) Answer : (3)

**Solution:**

Collenchyma provides mechanical support to the growing parts of plants such as young stem and petiole of leaf.

(101) Answer : (3)

**Solution:**

In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells.

When they are flaccid due to water stress, they make the leaves curl inwards to minimise water loss.

Bulliform cell are not primarily involved in photosynthesis.

(102) Answer : (3)

**Solution:**

Axillary buds are found in the axil of leaves and are capable of forming a flower or a tendril.

(103) Answer : (1)

**Solution:**

ER and Golgi apparatus synthesize and modify the proteins respectively.

(104) Answer : (1)

**Solution:**

The ribosomes of chloroplast are 70S type and made up of 50S and 30S subunits and are smaller than cytoplasmic ribosomes.

(105) Answer : (2)

**Solution:**

(i) Centromere – Primary constriction

(ii) Satellite – Small fragment due to secondary constriction

(iii) Kinetochore – Disc shaped structure

(iv) Histone – Basic protein

(106) Answer : (2)

**Solution:**

Blood has fibreless intercellular substance.

The other connective tissues (adipose and areolar) contain fibroblasts and fibres.

(107) Answer : (4)

**Solution:**

Carotenoids and anthocyanins are the secondary metabolites that belong to the category of pigments. Cellulose is a polymeric substance usually found in the cell wall of plants.

Ribose is a simple sugar and is a primary metabolite.

(108) Answer : (1)

**Solution:**

The entire body of *Periplaneta americana* is covered by a hard chitinous exoskeleton.

(109) Answer : (2)

**Solution:**

The nasal chamber opens into the pharynx, a portion of which is the common passage for food and air.

(110) Answer : (2)

**Solution:**

In pulmonary circulation deoxygenated blood moves from the heart to lungs for oxygenation and then returns the oxygenated blood back to the heart.

A unique vascular connection exists between the digestive tract and liver which is called the hepatic portal system.

Renal portal system is absent in humans.

(111) Answer : (1)

**Solution:**

• Percentage of proteins in the total cellular mass is 10-15%.

• The bond present between adjacent nucleotides is phosphodiester bond.

• Monosaccharides are joined together by glycosidic bonds.

(112) Answer : (3)

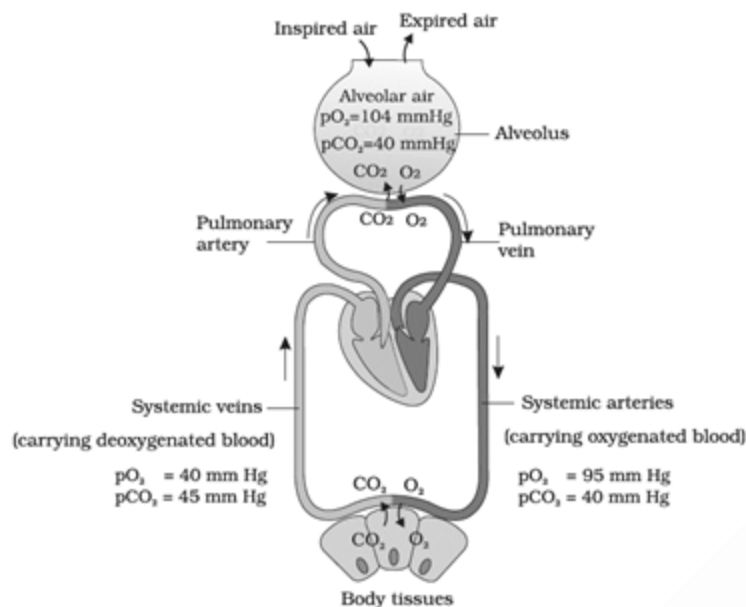
**Solution:**

Proteins are heteropolymers of amino acids.

Cellulose, starch and glycogen are homopolysaccharides.

(113) Answer : (3)

**Solution:**



(114) Answer : (3)

**Solution:**

The female reproductive system of cockroach consists of two large ovaries, lying laterally in the 2<sup>nd</sup> -6<sup>th</sup> abdominal segments.

(115) Answer : (3)

**Solution:**

After menopause, low levels of estrogen is a common cause of osteoporosis which leads to fracture of bones.

(116) Answer : (2)

**Solution:**

Ribosomes are common to both prokaryotic and eukaryotic cells.

(117) Answer : (3)

**Solution:**

The phenomenon, where extinction of the host fish leads to the extinction of unique assemblage of parasites is called co-extinction.

(118) Answer : (4)

**Solution:**

Mitosis is an equational division and it is seen in diploid cells as well as in haploid cells of lower plants and in some social insects.

(119) Answer : (2)

**Solution:**

Nucleus, Golgi complex and ER disappear by the end of prophase.

(120) Answer : (1)

**Solution:**

G<sub>1</sub> phase corresponds to the interval between mitosis and initiation of DNA replication.

(121) Answer : (3)

**Solution:**

Chlorophyll *b*, xanthophylls and carotenoids are accessory pigments. They enable a wide range of wavelength of incoming light to be utilized for photosynthesis. Carotenoids protect chlorophyll *a* from photo-oxidation.

(122) Answer : (4)

**Solution:**

C<sub>4</sub> plants show higher productivity and higher yield as compared to the C<sub>3</sub> plants.

(123) Answer : (3)

**Solution:**

Aerobic respiration takes place in aerobic prokaryotes too.

**(124) Answer :** (3)

**Solution:**

Substrate level phosphorylation takes place during conversion of PEP to pyruvic acid.

**(125) Answer :** (2)

**Solution:**

Developmental heterophylly is exhibited by cotton, coriander and larkspur. In such plants, the leaves of the juvenile plant are different in shape from those in mature plants. On the other hand, difference in shapes of leaves produced in air and those produced in water, in buttercup, represents environmental heterophylly.

**(126) Answer :** (4)

**Solution:**

Both kinetin and NAA are synthetic PGRs.

**(127) Answer :** (1)

**Solution:**

The period for which pollen grains remain viable is highly variable and to some extent depends on the prevailing temperature and humidity.

**(128) Answer :** (2)

**Solution:**

In monocots, the radicle and root cap is enclosed in a sheath called coleorhiza.

**(129) Answer :** (4)

**Solution:**

Wind pollinated flowers have light and non sticky pollens, well exposed stamens and large feathery stigma.

**(130) Answer :** (1)

**Solution:**

A true-breeding line is one that shows stable trait inheritance and expression for several generations. In pea plants, homozygous tall and dwarf traits are of pure line.

**(131) Answer :** (1)

**Solution:**

Sarcoplasmic reticulum is the store house of  $\text{Ca}^{2+}$ .

**(132) Answer :** (1)

**Solution:**

6 - 8 percent of plasma is made up of proteins and monocytes make up 6 - 8 percent of the total WBCs.

Erythrocytes, leucocytes and platelets are collectively called formed elements and they constitute nearly 45 per cent of the blood. 90-92 per cent of plasma is water. Neutrophils are the most abundant cells (60-65 per cent) of the total WBCs.

**(133) Answer :** (2)

**Solution:**

The channels present on axonal membrane are impermeable to negatively charged proteins but selectively permeable to ions.

**(134) Answer :** (1)

**Solution:**

On constriction of blood vessels, there is an increase in blood pressure which increases glomerular blood flow and thereby GFR.

**(135) Answer :** (4)

**Solution:**

Cortisol, testosterone, erythropoietin and thyroxine stimulate RBC production.

**(136) Answer :** (1)

**Solution:**

The body of hemichordates is differentiated into three regions:

Anteriormost – Proboscis

Middle part – Collar

Posteriormost – Trunk

**(137) Answer :** (1)

**Solution:**

During nerve impulse conduction, the rise in the stimulus-induced permeability to  $\text{Na}^+$  is short-lived, which is quickly followed by the rise in permeability to  $\text{K}^+$ . Within a fraction of second,  $\text{K}^+$  diffuses outside the membrane and restores the



resting potential of the membrane at the site of excitation and the excited nerve fibre becomes once more responsive to further stimulation.

(138) Answer : (4)

**Solution:**

Adaptive ability is inherited and it has genetic basis. According to Thomas Malthus, population size will grow exponentially if everybody reproduced maximally but in reality populations are stable in size because nature keeps control and resources are limited.

(139) Answer : (3)

**Solution:**

Branching descent and natural selection are the two key concepts of Darwinian theory of evolution.

(140) Answer : (1)

**Solution:**

The pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testis in males and ovary in females) are the organised endocrine glands in our body. ANF is secreted by the atrial wall of our heart. CCK is secreted by the GIT. Erythropoietin is secreted by the kidneys.

(141) Answer : (1)

**Solution:**

Evolution for Darwin was gradual while deVries believed mutation causes speciation. Saltation is a single step large mutation.

(142) Answer : (4)

**Solution:**

Both cortical and juxta-medullary nephrons form the filtrate. Peritubular capillaries are present in both.

In majority of nephrons, the loop of Henle is too short and extends only very little into the medulla. Such nephrons are called cortical nephrons.

In some of the nephrons, the loop of Henle is very long and runs deep into the medulla. These nephrons are called juxta-medullary nephrons.

Vasa recta is absent or highly reduced in cortical nephrons.

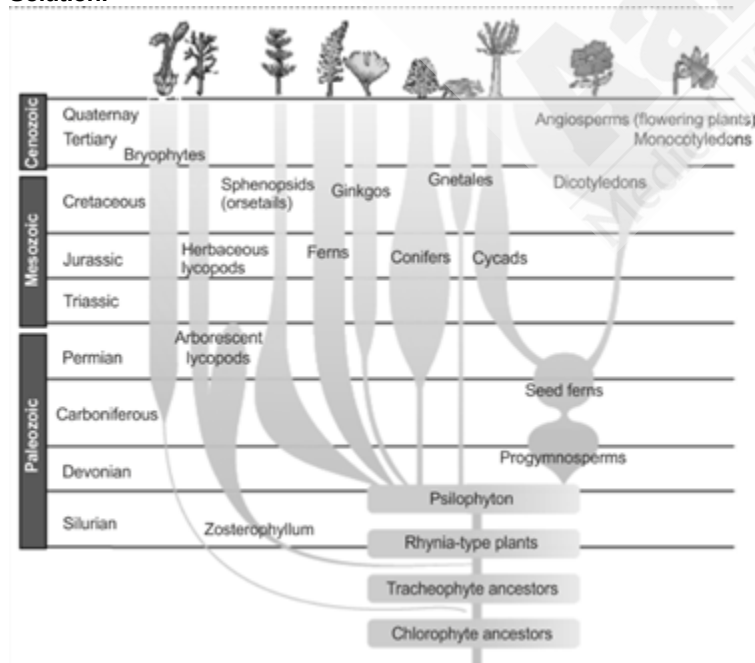
(143) Answer : (1)

**Solution:**

In agarose gel electrophoresis, DNA molecules are separated on the basis of size. The smaller the fragment, farther it moves away from the negative electrode (cathode).

(144) Answer : (3)

**Solution:**



(145) Answer : (2)

**Solution:**

Lymph nodes, tonsils, spleen, etc., are secondary lymphoid organs. These provide the site for interaction of lymphocytes with the antigen. Bone marrow is a primary lymphoid organ.

**(146) Answer :** (4)

**Solution:**

Frogs and humans have thyroid as well as parathyroid glands. Paired cerebral hemispheres and hepatic portal system are also present in both. Hind brain of frogs lack pons.

**(147) Answer :** (2)

**Solution:**

Bioprocess engineering: Maintenance of sterile (microbial contamination-free) ambience in chemical engineering processes to enable growth of only the desired microbe/eukaryotic cell in large quantities for the manufacture of biotechnological products like antibiotics, vaccines, enzymes, etc.

**(148) Answer :** (1)

**Solution:**



Increase in  $\text{Ca}^{+2}$  level leads to the binding of calcium with a subunit of troponin on actin filaments and thereby remove the masking of active sites for myosin.

**(149) Answer :** (2)

**Solution:**

Ova from the wife/donor (female) and sperms from the husband/donor (male) are collected and are induced to form the zygote under simulated conditions in the laboratory.

Then, the zygote or early embryos (with upto 8 blastomeres) can be transferred into the fallopian tube, which is called ZIFT (Zygote Intra Fallopian Transfer).

**(150) Answer :** (3)

**Solution:**

The exchange of materials between blood and tissue fluid, primarily oxygen, nutrients, and waste products, occurs across the thin walls of capillaries, facilitated by diffusion, filtration, and osmosis.

**(151) Answer :** (2)

**Solution:**

Allergy is due to the release of chemicals like histamine and serotonin from the mast cells.

**(152) Answer :** (2)

**Solution:**

Gastrin acts on the gastric glands and stimulates the secretion of hydrochloric acid and pepsinogen.

Secretin acts on the exocrine pancreas and stimulates the secretion of water and bicarbonate ions.

CCK acts on both pancreas and gall bladder and stimulates the secretion of pancreatic enzymes and bile juice, respectively.

GIP inhibits gastric secretion and motility.

**(153) Answer :** (3)

**Solution:**

The Indian Government has set up organisations such as GEAC (Genetic Engineering Approval Committee), which makes decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services.

**(154) Answer :** (3)

**Solution:**

Digestion in porifers (sponges) is intracellular, meaning food particles are digested within the cells themselves.

**(155) Answer :** (2)

**Solution:**

Presence of insert results into the insertional inactivation of the  $\beta$ -galactosidase and thus, the colonies do not produce any colour. These colonies are identified as recombinant colonies.

**(156) Answer :** (3)

**Solution:**

Less number of chromosomes help to identify the specific ones w.r.t. characters. Sexual reproduction is required to study the inheritance of characters through chromosomes.

**(157) Answer :** (2)

**Solution:**

Sickle cell anemia is caused by the substitution of glutamic acid by valine.

**(158) Answer :** (2)

**Solution:**

In co-dominance, the  $F_1$  generation resembles both the parents.

(159) Answer : (1)

**Solution:**

The main enzyme for DNA replication is DNA dependent DNA polymerase.

(160) Answer : (2)

**Solution:**

In Griffith experiment, heat killed strain of pathogenic *Streptococcus pneumoniae* makes the non-pathogenic strain into pathogenic one.

(161) Answer : (3)

**Solution:**

Central dogma in molecular biology was proposed by Francis Crick.

(162) Answer : (1)

**Solution:**

The correct sequence of the steps in DNA fingerprinting is (E) → (B) → (C) → (D) → (A)

(163) Answer : (2)

**Solution:**

tRNA is non-catalytic in nature.

(164) Answer : (2)

**Solution:**

*Trichoderma* species are free-living fungi that are very common in the root ecosystems. They are effective biocontrol agents of several plant pathogens.

(165) Answer : (2)

**Solution:**

One of the main objectives of Ganga action plan and Yamuna action plan was to build sewage treatment plants (STPs), so that treated sewage could be discharged in the rivers.

(166) Answer : (2)

**Solution:**

Sea anemone and clown fish show commensalism.

(167) Answer : (3)

**Solution:**

Total number count is not applicable to calculate bacterial population density.

(168) Answer : (3)

**Solution:**

The integral form of exponential growth equation is  $N_t = N_0 e^{rt}$

Here,  $N_t$  = population density after time  $t$

$N_0$  = population density at time zero

$e = 2.71828$

(169) Answer : (3)

**Solution:**

In sea ecosystem, the biomass of producers (phytoplankton) is less than the biomass of consumers.

(170) Answer : (4)

**Solution:**

Kangaroo rat is able to meet all its water requirements through internal fat oxidation and it has the ability to concentrate its urine.

(171) Answer : (1)

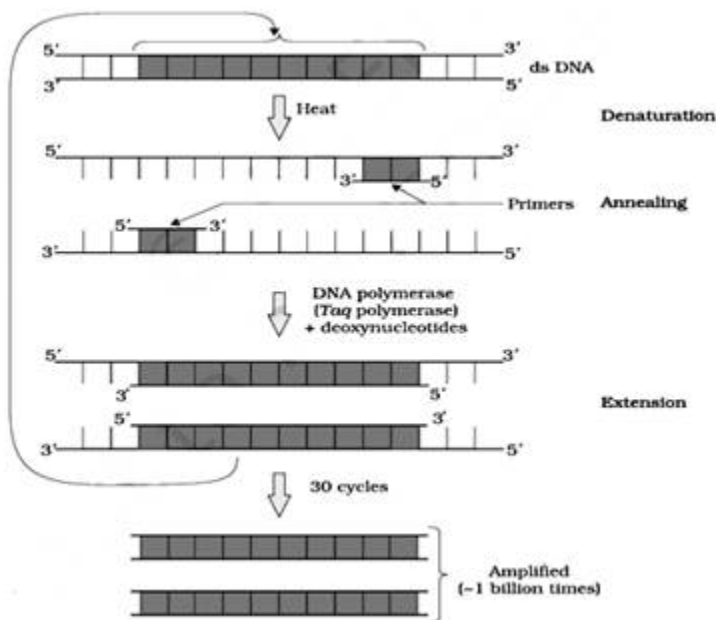
**Solution:**

Basmati rice is distinct for its unique aroma and flavour and 27 documented varieties of Basmati are grown in India.

Rice is an important food grain, the presence of which goes back thousands of years in Asia's agricultural history. There are an estimated 200,000 varieties of rice in India.

(172) Answer : (1)

**Solution:**



(173) Answer : (2)

**Solution:**

Enzymes catalysing the hydrolysis of ester, ether, peptide, glycosidic, C-C, C-halide or P-N bonds are called hydrolases.

(174) Answer : (3)

**Solution:**

Oxytocin is known as the milk-let down hormone and it also participates in foetal-ejection reflex. Oxytocin is formed in the hypothalamus.

(175) Answer : (2)

**Solution:**

5'-G↓AATTC-3'

3'-CTTAA↑G-5'

EcoRI cuts the DNA between bases G and A.

(176) Answer : (2)

**Solution:**

Using conventional methods of diagnosis (serum and urine analysis, etc.) early detection is not possible.

Recombinant DNA technology, Polymerase Chain Reaction (PCR) and Enzyme Linked Immuno-Sorbent Assay (ELISA) are some of the techniques that serve the purpose of early diagnosis.

(177) Answer : (1)

**Solution:**

Oral administration of small doses of either progestogens or progestogen–estrogen combinations is one of the contraceptive methods used by the females. They are used in the form of tablets and hence are popularly called the pills. They inhibit ovulation and implantation as well as alter the quality of cervical mucus to prevent/retard entry of sperms. IUDs increase phagocytosis of sperms within the uterus and the Cu ions released suppress sperm motility and the fertilising capacity of sperms.

(178) Answer : (1)

**Solution:**

The SAN (pacemaker) can generate the maximum number of action potentials, i.e.,  $70-75 \text{ min}^{-1}$ , and is responsible for initiating and maintaining the rhythmic contractile activity of the heart.

(179) Answer : (4)

**Solution:**

During pregnancy, the levels of other hormones like estrogens, progestogens, cortisol, prolactin, thyroxine, etc., are increased several folds in the maternal blood.

Increased production of these hormones is essential for supporting the foetal growth, metabolic changes in the mother and maintenance of pregnancy.

(180) Answer : (3)

**Solution:**

In a 28 days cycle, ovulation usually occurs around the 14<sup>th</sup> day, thus, LH surge occurs prior to ovulation *i.e.*, around 36 hours before ovulation.

